Class 16: Obtaining and processing SRA datasets on AWS

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Working with RNA-Seq data

> Q1. How would you check that these files with extension '.fastq' actually look like what we expect for a FASTQ file? You could try printing the first few lines to the shell standard output:

On AWS

$ head *.fastq

> Q2. How could you check the number of sequences in each file?

On AWS

$ grep -c "@SRR" *.fastq

> Q3. Check your answer with the bottom of the file using the tail command and also check the matching mate pair FASTQ file. Do these numbers match? If so why or why not?

On AWS

$ ls *.fastq

> Q4. Can you run kallisto to print out it's citation information?

On AWS

$ kallisto –cite

> Q5. Have a look at the TSV format versions of these files to understand their structure. What do you notice about these files contents?

On AWS

$ head *_quant/*.tsv

Downstream analysis

```r
library(tximport)
```

```r
folders <- dir(pattern = "SRR21568*")
samples <- sub("_quant", "", folders)
files <- file.path( folders, "abundance.h5" )
names(files) <- samples

txi.kallisto <- tximport(files, type = "kallisto", txOut = TRUE)
```

```
1 2 3 4
```

```r
head(txi.kallisto$counts)
```

```
                SRR2156848 SRR2156849 SRR2156850 SRR2156851
ENST00000539570          0          0    0.00000          0
ENST00000576455          0          0    2.62037          0
ENST00000510508          0          0    0.00000          0
ENST00000474471          0          1    1.00000          0
ENST00000381700          0          0    0.00000          0
ENST00000445946          0          0    0.00000          0
```

```r
colSums(txi.kallisto$counts)
```

```
SRR2156848 SRR2156849 SRR2156850 SRR2156851
   2563611    2600800    2372309    2111474
```

```r
sum(rowSums(txi.kallisto$counts)>0)
```

```
[1] 94561
```

```r
to.keep <- rowSums(txi.kallisto$counts) > 0
kset.nonzero <- txi.kallisto$counts[to.keep,]
```

```r
keep2 <- apply(kset.nonzero,1,sd)>0
x <- kset.nonzero[keep2,]
```

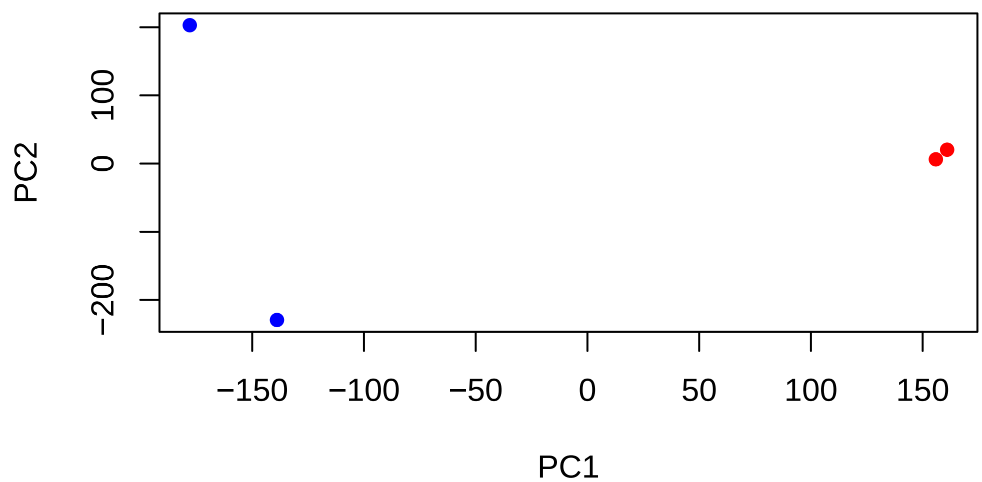
Principal Component Analysis

```
pca <- prcomp(t(x), scale=TRUE)
```

```
summary(pca)
```

```
Importance of components:
                            PC1      PC2      PC3   PC4
Standard deviation     183.6379 177.3605 171.3020 1e+00
Proportion of Variance   0.3568   0.3328   0.3104 1e-05
Cumulative Proportion    0.3568   0.6895   1.0000 1e+00
```

```
plot(pca$x[,1], pca$x[,2],
     col=c("blue","blue","red","red"),
     xlab="PC1", ylab="PC2", pch=16)
```

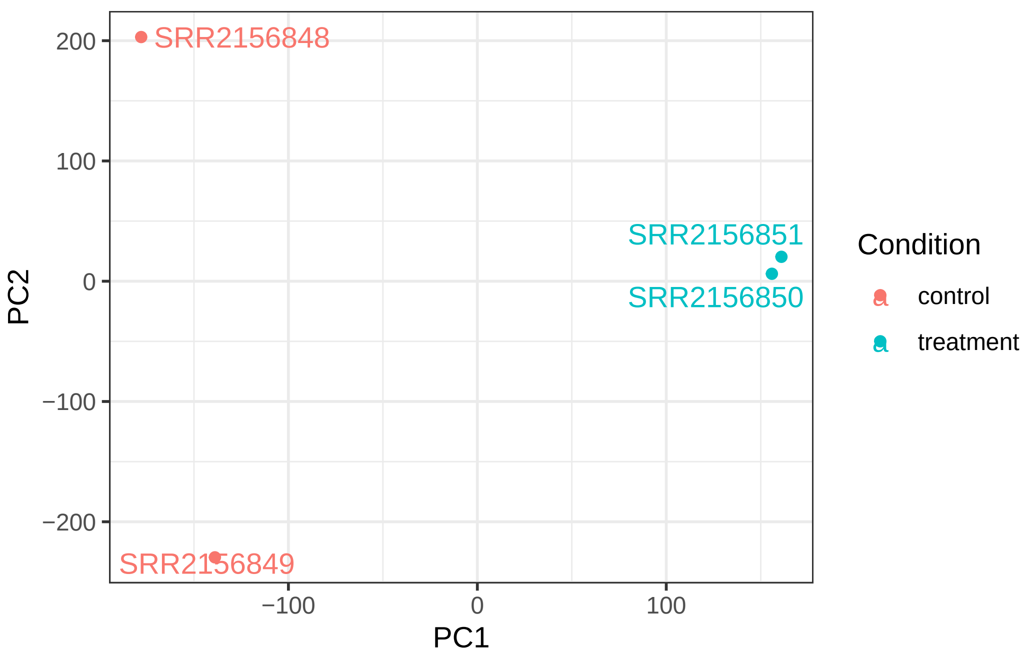


Q6. Use ggplot to make a similar figure of PC1 vs PC2 and a seperate figure PC1 vs PC3 and PC2 vs PC3.

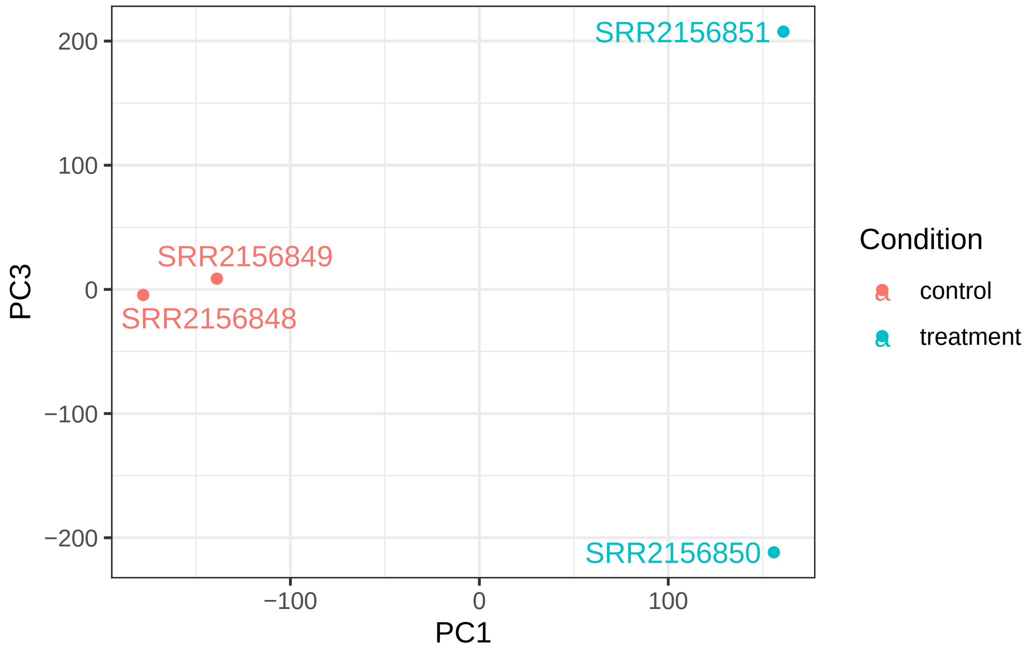
```r
library(ggplot2)
library(ggrepel)
```

```
Warning: package 'ggrepel' was built under R version 4.3.3
```

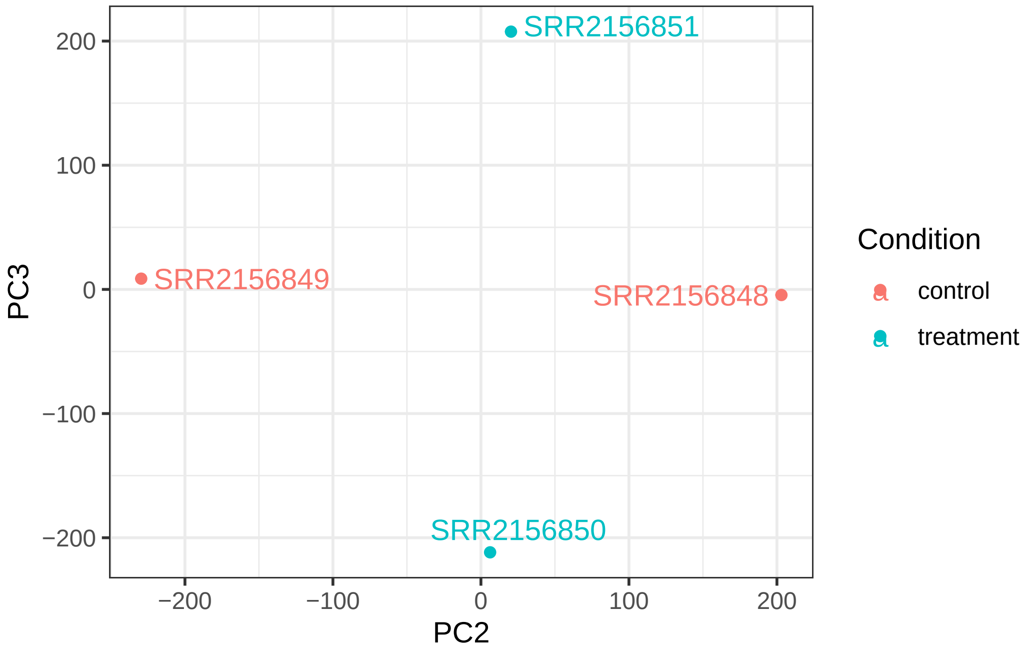
```r
colData <- data.frame(condition = factor(rep(c("control", "treatment"), each = 2)))
rownames(colData) <- colnames(txi.kallisto$counts)

y <- as.data.frame(pca$x)
y$Condition <- as.factor(colData$condition)

ggplot(y) +
  aes(PC1, PC2, col=Condition) +
  geom_point() +
  geom_text_repel(label=rownames(y)) +
  theme_bw()
```




```r
ggplot(y) +
  aes(PC1, PC3, col=Condition) +
  geom_point() +
  geom_text_repel(label=rownames(y)) +
  theme_bw()
```




```
ggplot(y) +
  aes(PC2, PC3, col=Condition) +
  geom_point() +
  geom_text_repel(label=rownames(y)) +
  theme_bw()
```


SRR2156851
SRR2156849
SRR2156848
SRR2156850
200
100
0
−100
−200
PC3
−200
−100
0
100
200
PC2
Condition
control
treatment